



Assignment Report

Course Details

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Submitted To

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Table of Contents

1. Introduction	2
2. Network Topology Overview	2
2.1 Overall Architecture.....	2
2.4 DHCP Server Configuration	12
2.6 DSN Server Configuration.....	20
2.7 Static Routing Configuration.....	20
3. Experimental Results	23
3.1 Ping Command.....	23
3.2 Ping Test Results	25
4. Conclusion	26

1. Introduction

This assignment focuses on the design and configuration of a complete network based on a given topology using four routers. The network has been implemented in Cisco Packet Tracer by following proper network planning techniques, including logical design, correct IP addressing and structured connectivity between different networks.

While designing the network, I use only Class A for LAN and Class B IP addresses for addressed through proper IP addressing and subnetting to avoid conflicts and ensure smooth communication. The design connects multiple local area networks through router to router links, forming a complete and connected system.

The network is configured in a way that hosts in each LAN can obtain IP addresses automatically using DHCP, where required by the topology. At the same time, static routing is applied on all routers to enable end to end communication between all networks. The overall implementation aims to achieve reliable connectivity across the entire network.

Finally, the correctness of the network configuration is verified through successful ping tests between selected end devices, confirming that the network operates as expected.

2. Network Topology Overview

2.1 Overall Architecture

The topology consists of:

- 4 Routers connected using serial DCE
- Switches to connect multiple end devices in each LAN
- PCs and Servers as end devices
- Separate DHCP services for each LAN

Each router connects two networks:

- LAN network (via FastEthernet)
- WAN links (via Serial DCE)

➤ Network Design

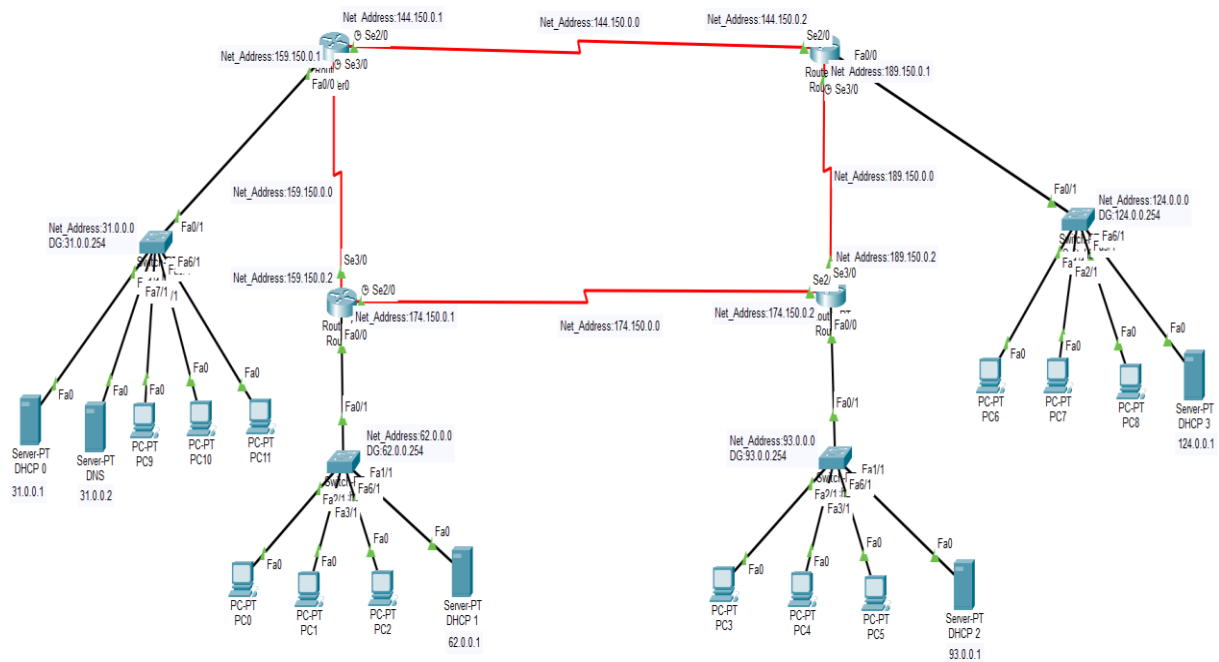


Figure 1: Complete Network Topology Designed Using Four Routers in Cisco Packet Tracer

2.2 IP Addressing Scheme

Only Class A and Class B IP addresses are used in this network.

➤ LAN Networks

This table shows the LAN wise network addressing configuration. Each LAN has a unique network address, a router-based default gateway and a dedicated DHCP server.

LAN	Network Address	Default Gateway	DHCP Server
LAN 1	31.0.0.0	31.0.0.254	31.0.0.1
LAN 2	62.0.0.0	62.0.0.254	62.0.0.1
LAN 3	93.0.0.0	93.0.0.254	93.0.0.1
LAN 4	124.0.0.0	124.0.0.254	124.0.0.1

Table 1: LAN Network Addressing and DHCP Configuration

➤ WAN (Serial DCE) Networks

This table represents the point to point network links between routers. Each router to router link is assigned a unique network address to ensure proper routing. The router interfaces use sequential IP addresses for communication between connected routers, enabling reliable and efficient data transfer across the network.

Link	Network Address	Router Interfaces
R0 to R2	159.150.0.0	159.150.0.1 & 159.150.0.2
R0 to R1	144.150.0.0	144.150.0.1 & 144.150.0.2
R1 to R3	189.150.0.0	189.150.0.1 & 189.150.0.2
R2 to R3	174.150.0.0	174.150.0.1 & 174.150.0.2

Table 2: WAN Link Addressing Between Routers

➤ PC IP Address Assignment (DHCPB based)

This table shows the IP addressing schemes for all PCs connected to the LANs in the network. Each LAN has its own unique network address and each PC is assigned a static IP within that network. The subnet mask is set to 255.0.0.0, allowing many hosts per LAN. The default gateway for each LAN points to the router interface connected to that LAN, enabling PCs to communicate with other LANs and the wider network.

PC Name	IP Address	Subnet Mask	Default Gateway
PC0	62.0.0.10	255.0.0.0	62.0.0.254
PC1	62.0.0.11	255.0.0.0	62.0.0.254
PC2	62.0.0.12	255.0.0.0	62.0.0.254
PC3	93.0.0.10	255.0.0.0	93.0.0.254
PC4	93.0.0.11	255.0.0.0	93.0.0.254
PC5	93.0.0.12	255.0.0.0	93.0.0.254
PC6	124.0.0.10	255.0.0.0	124.0.0.254
PC7	124.0.0.11	255.0.0.0	124.0.0.254
PC8	124.0.0.12	255.0.0.0	124.0.0.254
PC9	31.0.0.10	255.0.0.0	31.0.0.254
PC10	31.0.0.11	255.0.0.0	31.0.0.254
PC11	31.0.0.12	255.0.0.0	31.0.0.254

Table 3: LAN IP Addressing and PC Configuration

2.3 Router Interface Configuration

The first step of the network implementation involved configuring all router interfaces with static IP addresses. FastEthernet interfaces were configured to serve as default gateways for their respective LANs, while Serial interfaces were assigned Class B IP addresses to enable inter-router communication. Class A addresses were used for connections between switches and routers over copper straight through cables. Serial DCE interfaces were configured with appropriate clock rates to ensure proper synchronization. This initial configuration established the foundational connectivity required for higher-level network services.

➤ For Router 0:

se 2/0

Router>en

```
Router#config t
Router(config)#int se 2/0
Router(config-if)#ip address 144.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 144.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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Se 3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

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fa 0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 31.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```

Router(config)#int fa0/0
Router(config-if)#ip address 31.0.0.254 255.0.0.0
Router(config-if)#no shut

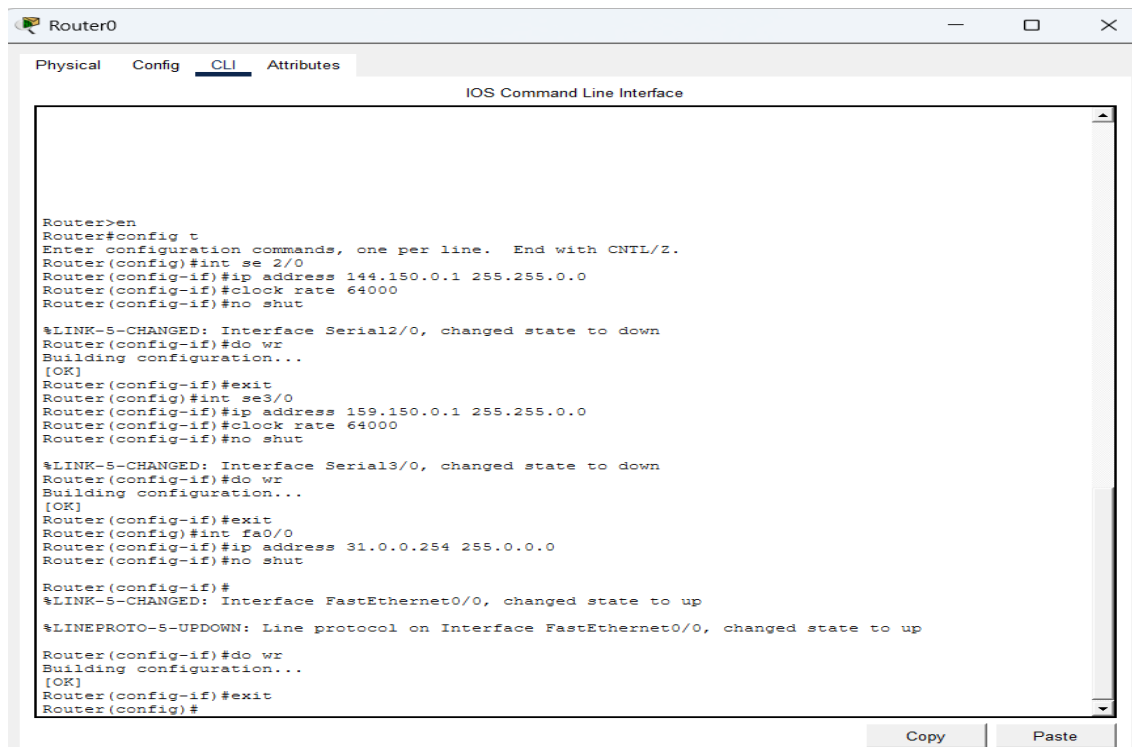
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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The screenshot shows a window titled 'Router0' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following commands and responses:

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 144.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 31.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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Figure 2: Router Interface Configuration (CLI) for Router 0

➤ For Router 1

se2/0

```

Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 144.150.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 144.150.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#

```

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Se 3/0

```

Router(config)#int se3/0
Router(config-if)#ip address 189.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router(config)#int se3/0
Router(config-if)#ip address 189.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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fa 0/0

```

Router(config)#int fa0/0
Router(config-if)#ip address 124.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router(config)#int fa0/0
Router(config-if)#ip address 124.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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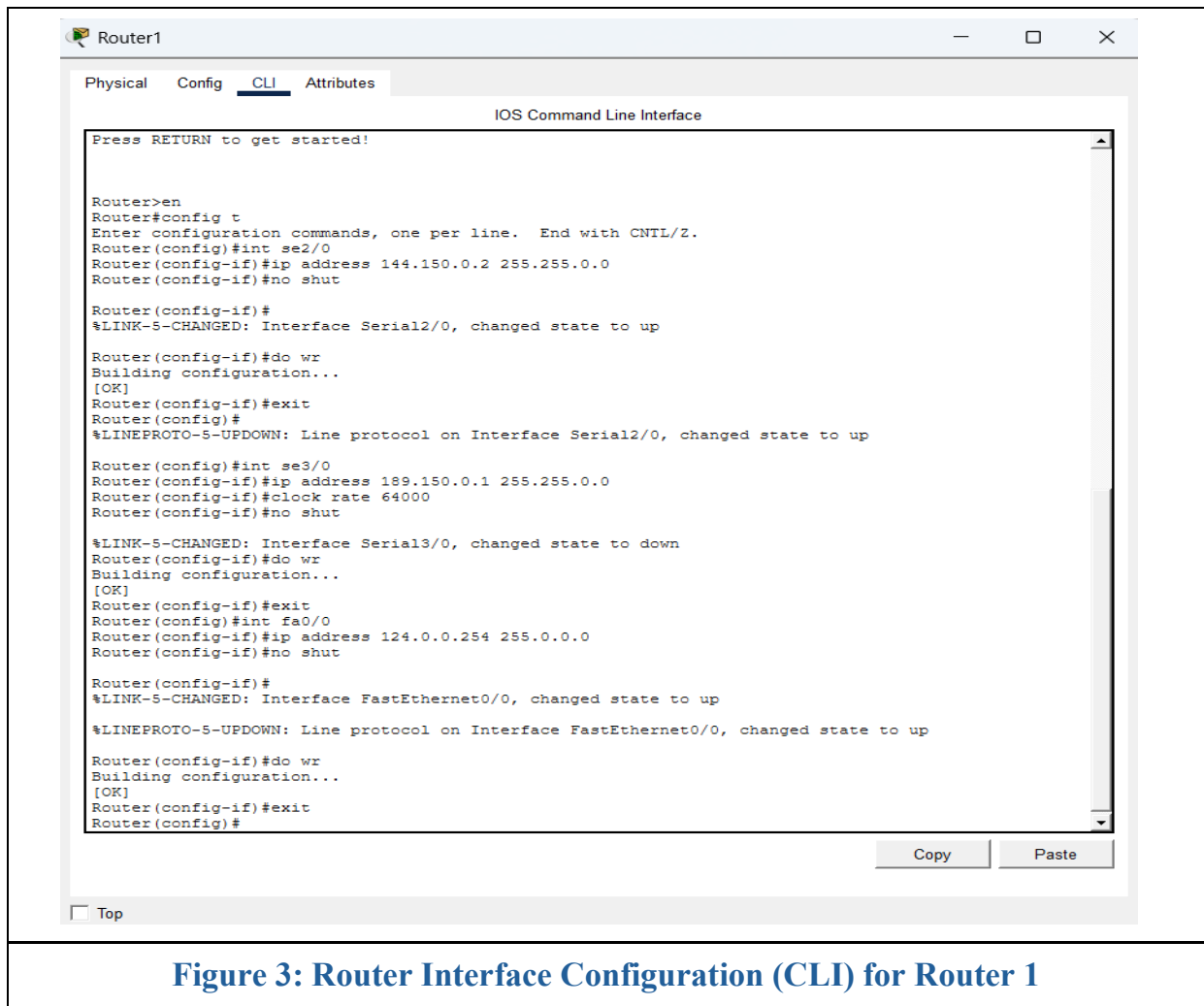


Figure 3: Router Interface Configuration (CLI) for Router 1

➤ For Router 2

se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 174.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 174.150.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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Se3/0

```

Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router(config)#int se3/0
Router(config-if)#ip address 159.150.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
Router(config)#

```

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fa0/0

```

Router(config)#int fa0/0
Router(config-if)#ip address 62.0.0.254 255.0.0.0
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router(config)#int fa0/0
Router(config-if)#ip address 62.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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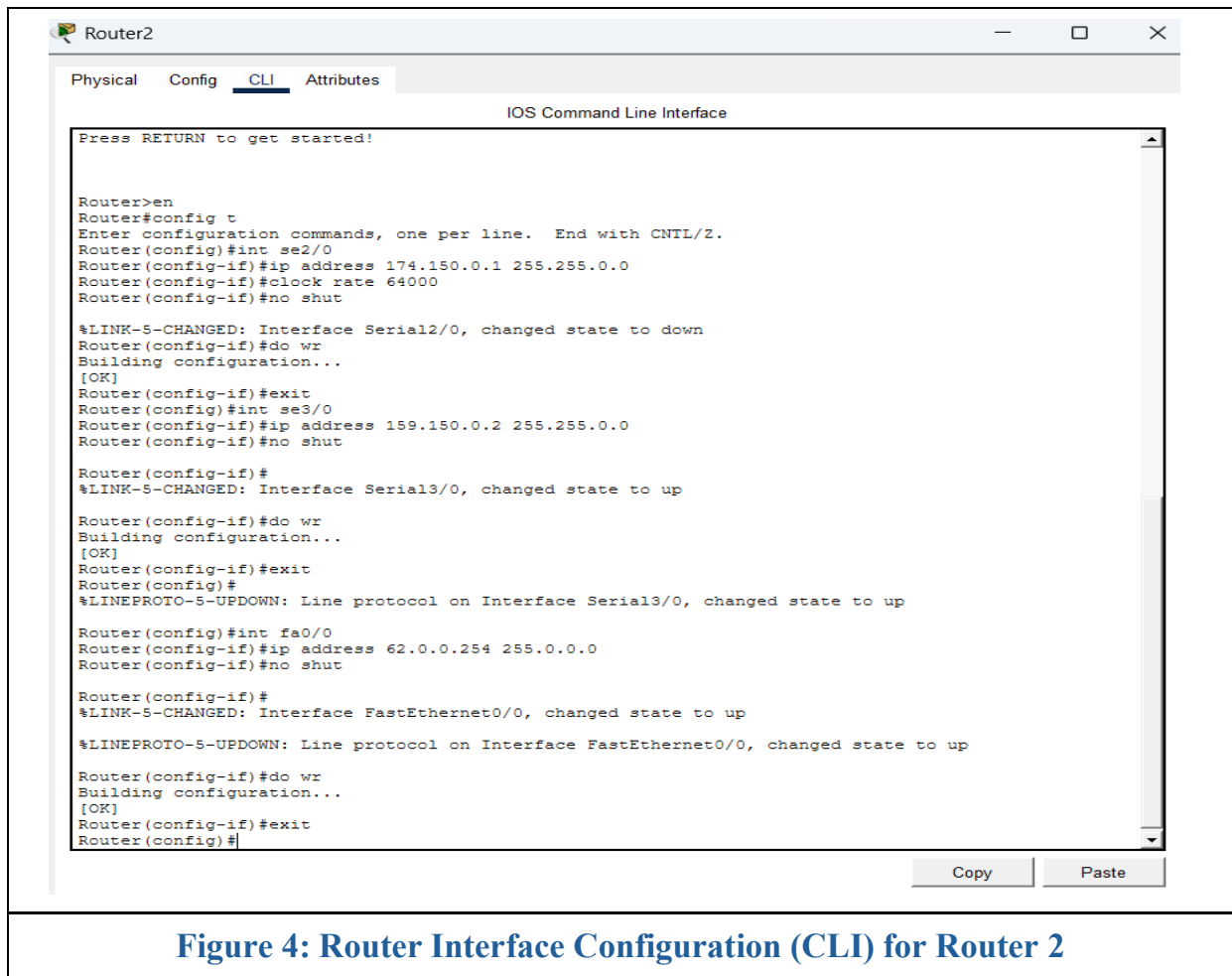


Figure 4: Router Interface Configuration (CLI) for Router 2

➤ For Router 3

Se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 174.150.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 174.150.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config)#

```

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fa0/0

```

Router(config)#int fa0/0
Router(config-if)#ip address 93.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router(config)#int fa0/0
Router(config-if)#ip address 93.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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Se 3/0

```

Router(config)#int se3/0
Router(config-if)#ip address 189.150.0.2 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

```

```

Router(config)#int se3/0
Router(config-if)#ip address 189.150.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#

```

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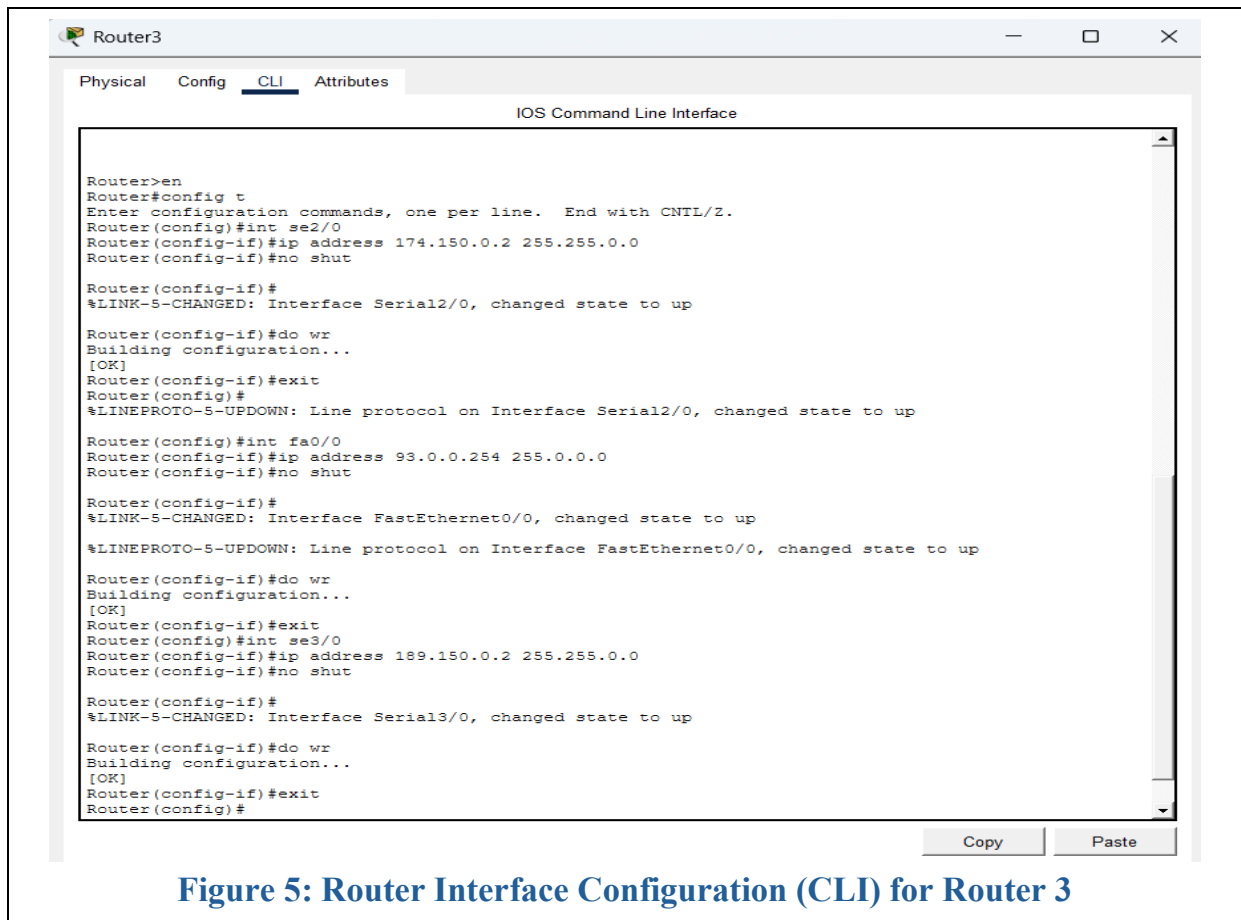


Figure 5: Router Interface Configuration (CLI) for Router 3

2.4 DHCP Server Configuration

Each LANs contains a dedicated DHCP server connected to the switch. The DHCP server is responsible for automatically assigning IP addresses to hosts within its respective LAN.

Each DHCP server was configured with the following parameters:

- Network address and subnet mask
- Default gateway (router LAN interface)
- DNS server IP address

All PCs were configured to obtain IP addresses automatically using DHCP. This distributed DHCP design reduces manual configuration, improves reliability and ensures efficient IP address management within each LANs

➤ DHCP 0

The screenshot shows the 'DHCP 0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the 'Static' radio button selected. The fields are filled with the following values:

Field	Value
IPv4 Address	31.0.0.1
Subnet Mask	255.0.0.0
Default Gateway	31.0.0.254
DNS Server	31.0.0.2

Figure 6: DHCP Configuration

The screenshot shows the 'DHCP 0' configuration window with the 'Services' tab selected. The 'DHCP' service is configured for the 'FastEthernet0' interface and is turned 'On'. The configuration details are as follows:

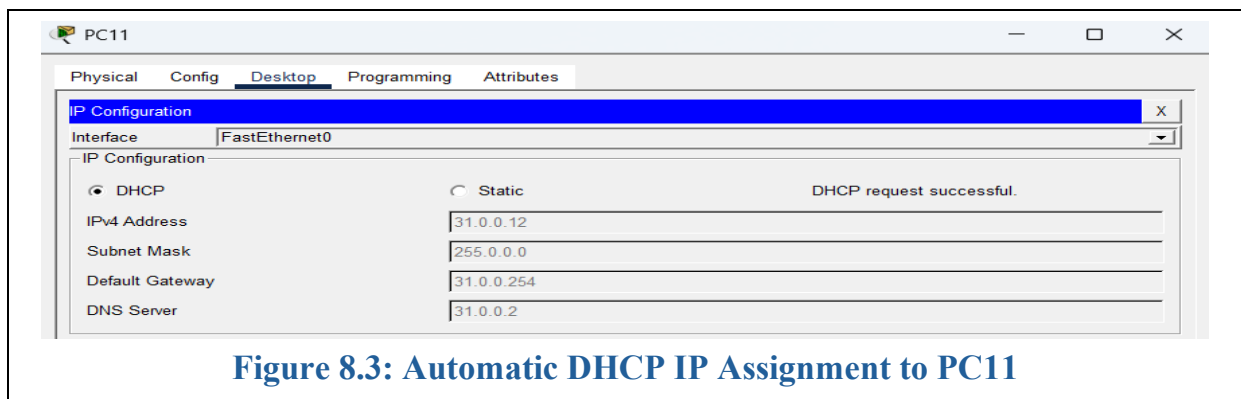
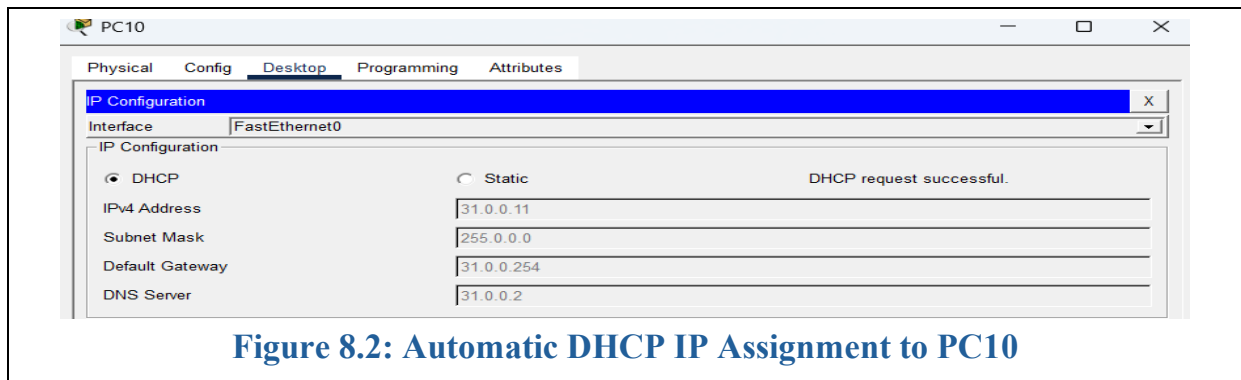
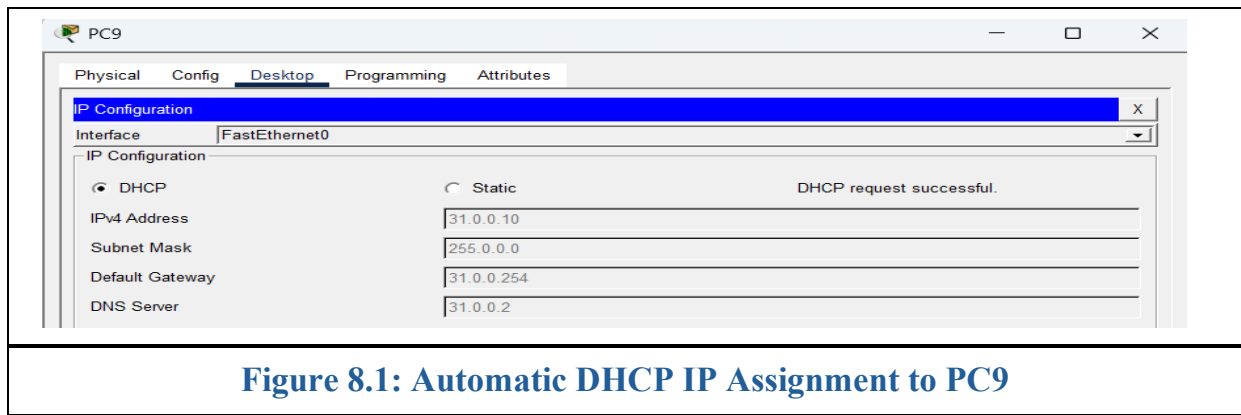
Field	Value
Interface	FastEthernet0
Service	On
Pool Name	serverPool
Default Gateway	31.0.0.254
DNS Server	31.0.0.2
Start IP Address	31.0.0.10
Subnet Mask	255.0.0.0
Maximum Number of Users	512
TFTP Server	0.0.0.0
WLC Address	0.0.0.0

Below the configuration fields is a table showing the active DHCP pool:

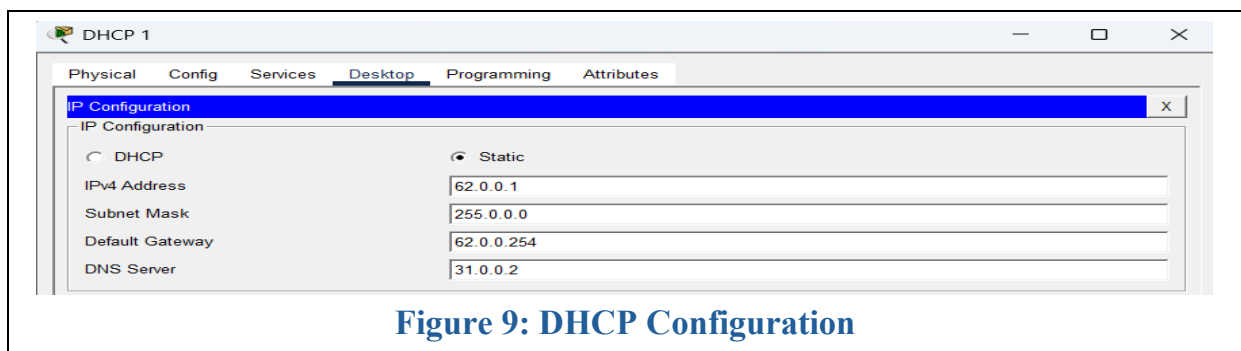
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	31.0.0.254	31.0.0.2	31.0.0.10	255.0.0.0	512	0.0.0.0	0.0.0.0

Figure 7: DHCP0 Service Running

➤ Automatic IP Assignment to PCs



➤ DHCP 1



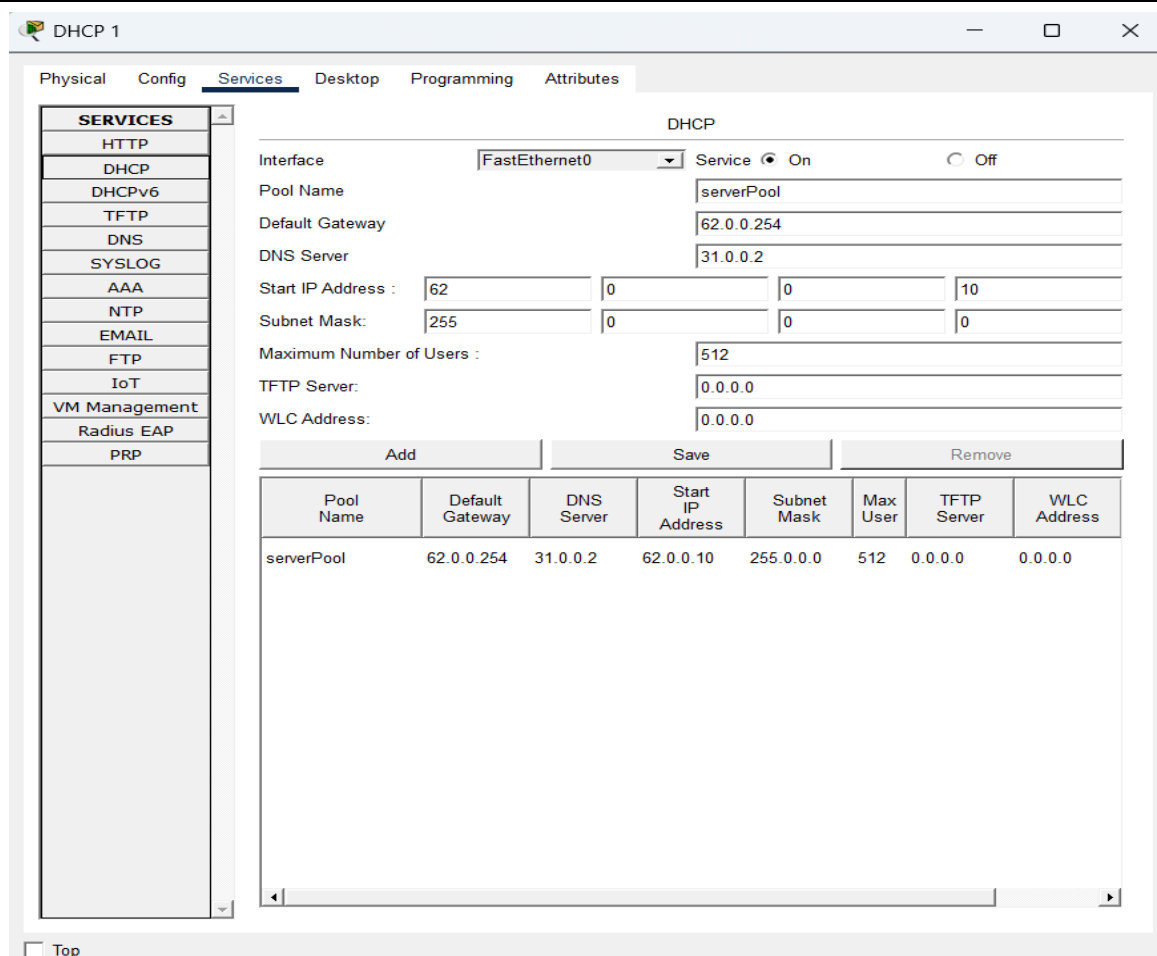


Figure 10: DHCP1 Service Running

➤ Automatic IP Assignment to PCs

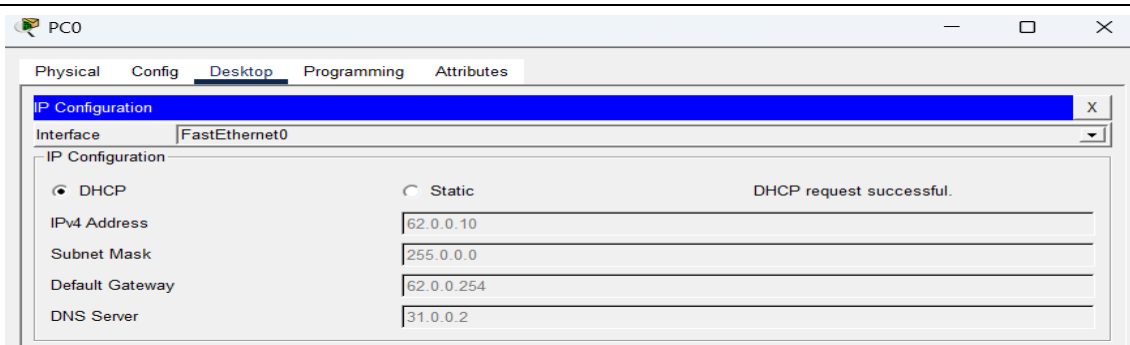


Figure 11.1: Automatic DHCP IP Assignment to PC 0

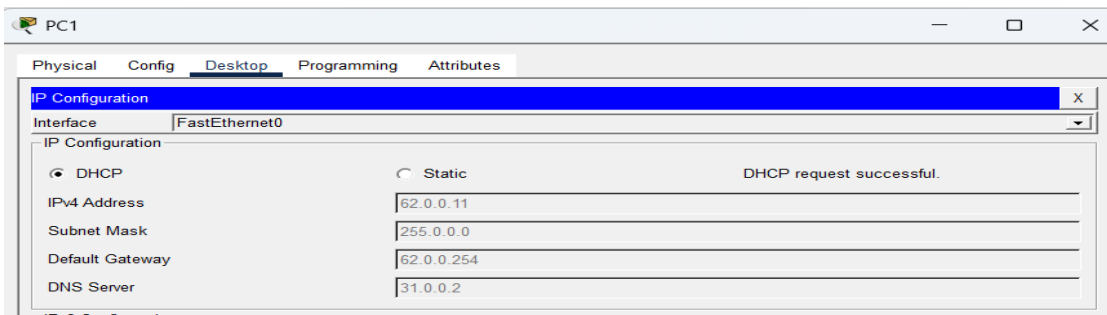


Figure 11.2: Automatic DHCP IP Assignment to PC 1

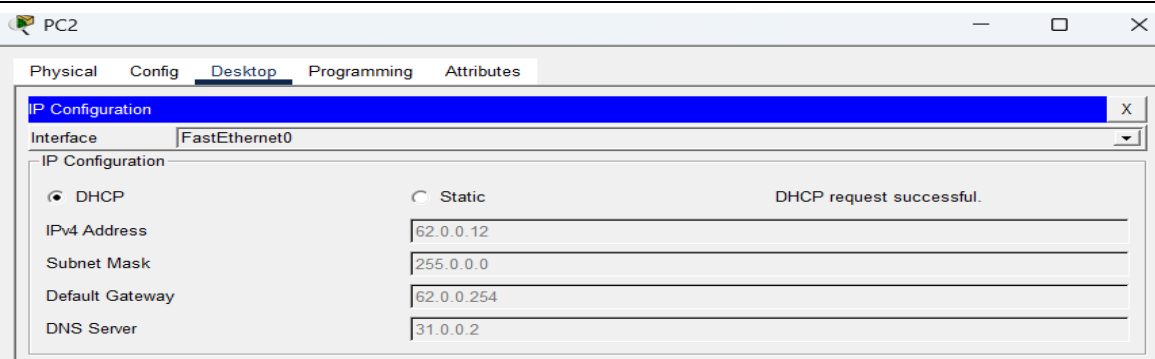


Figure 11.3: Automatic DHCP IP Assignment to PC 2

➤ DHCP 2

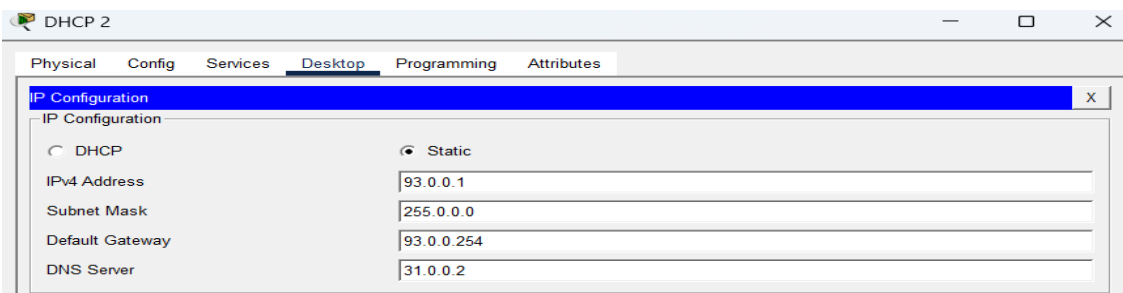


Figure 12: DHCP Configuration

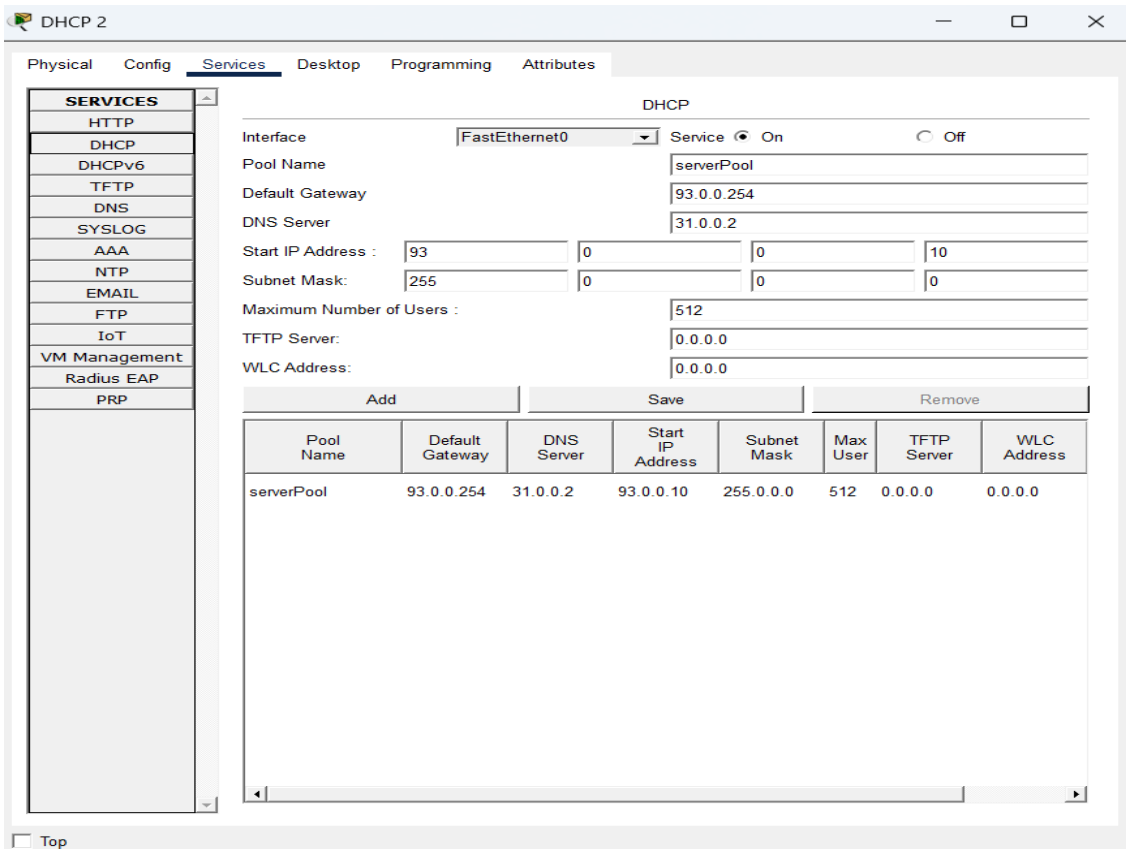


Figure 13: DHCP1 Service Running

➤ Automatic IP Assignment to PCs

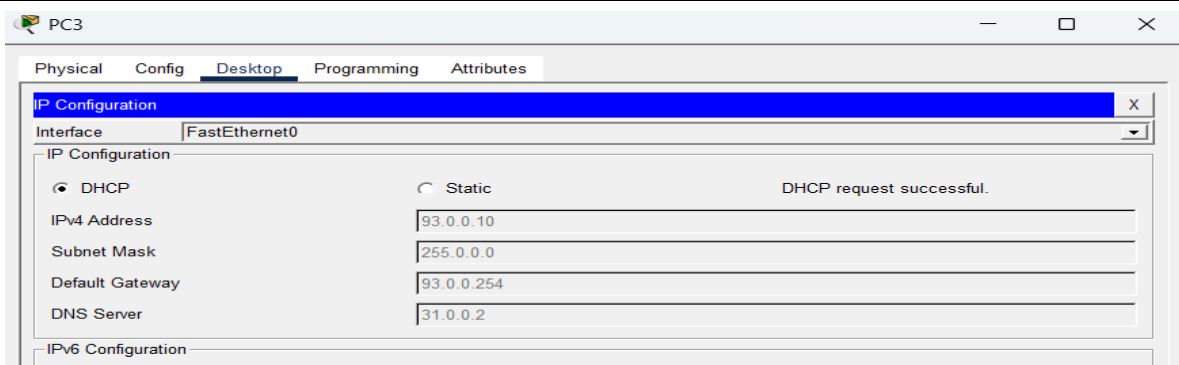
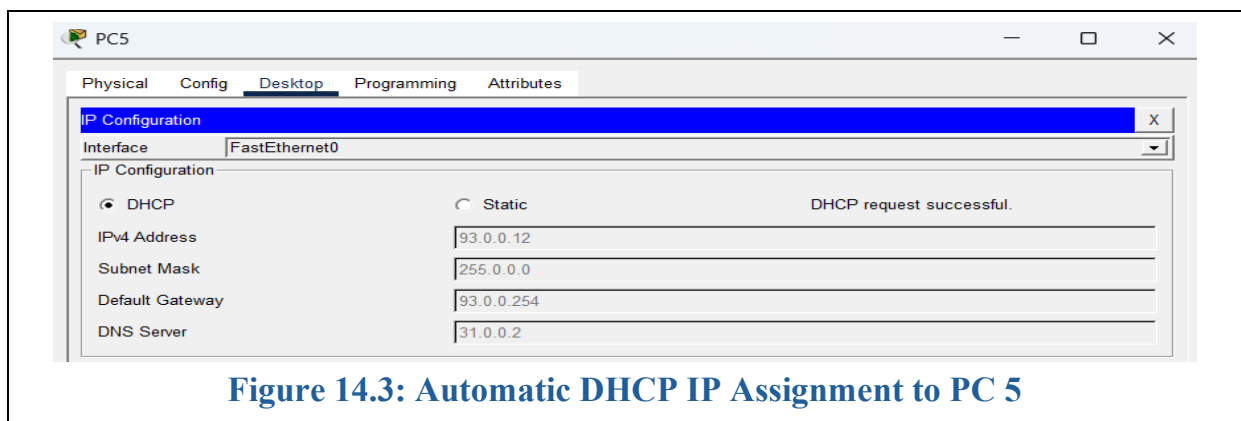
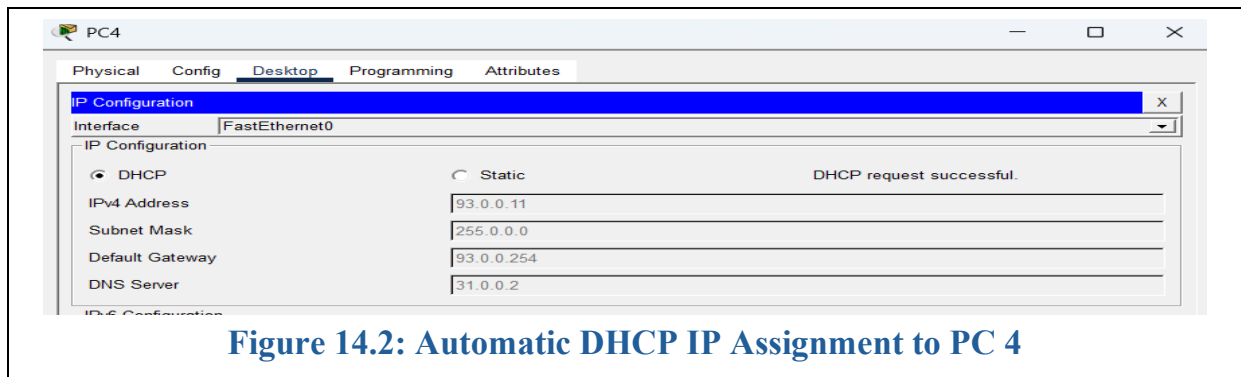
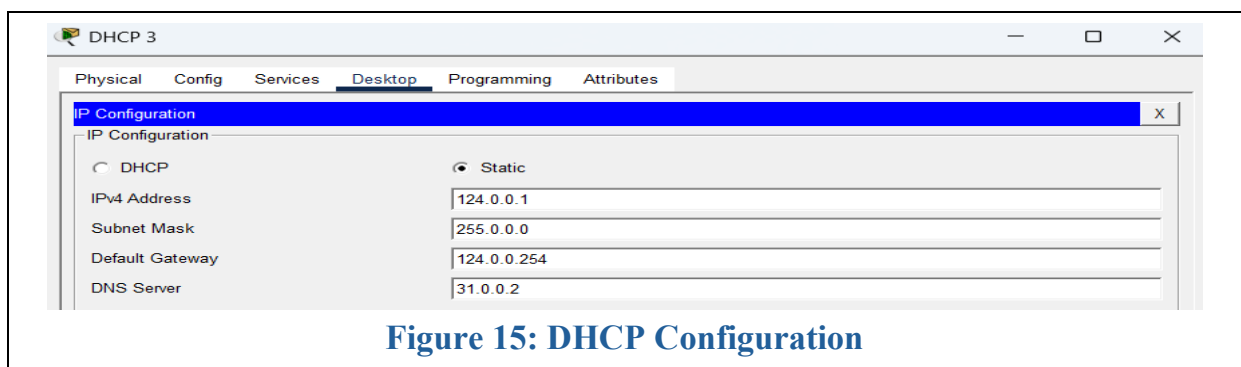
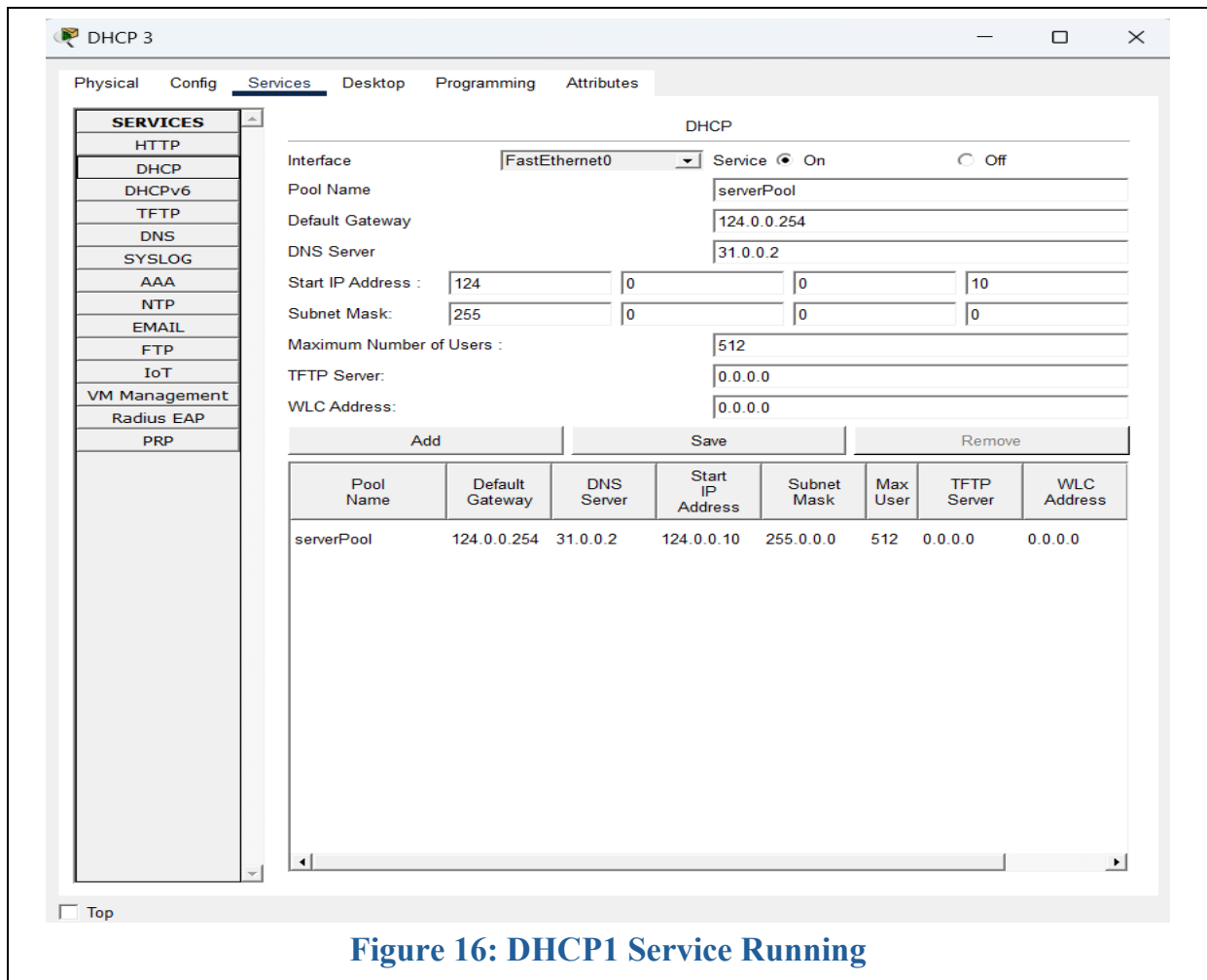


Figure 14.1: Automatic DHCP IP Assignment to PC 3

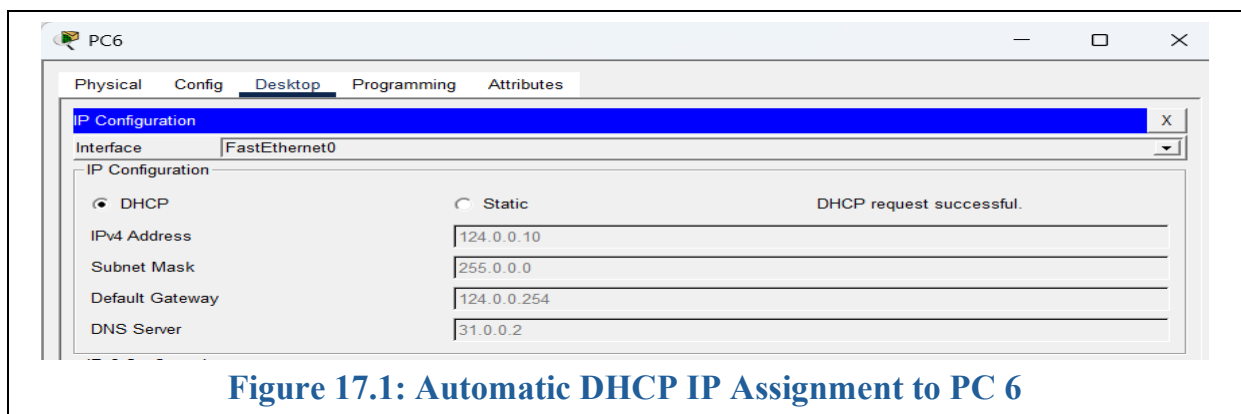


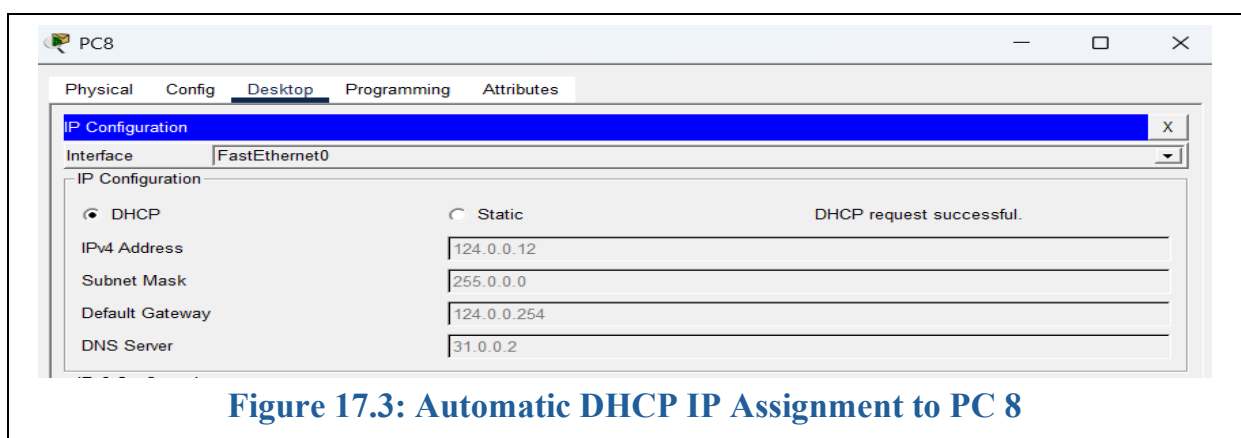
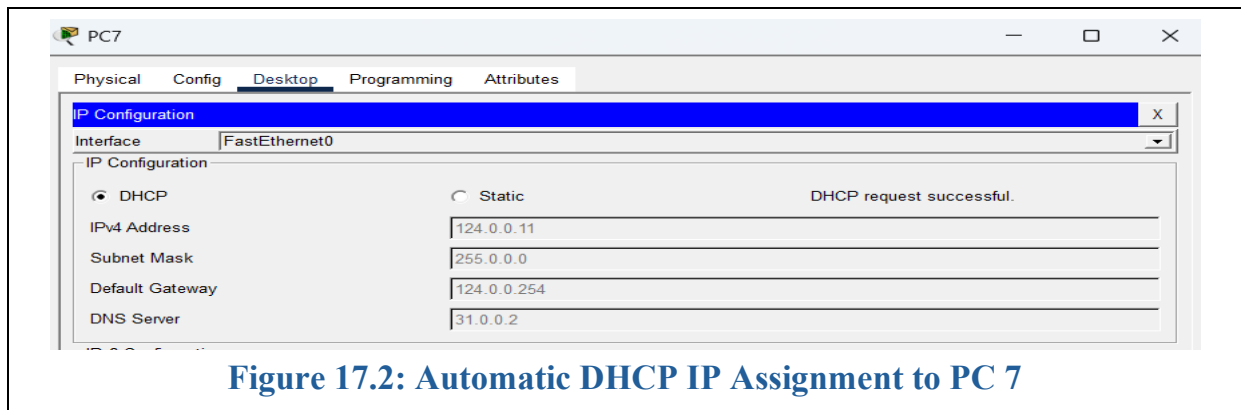
➤ DHCP 3





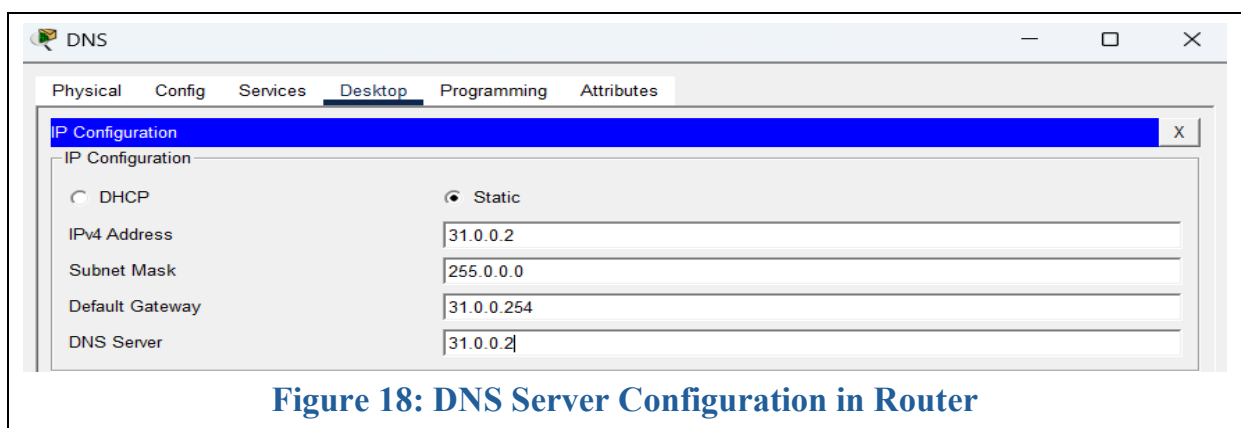
➤ Automatic IP Assignment to PCs





2.6 DSN Server Configuration

After completing DHCP server, a centralized DNS server was configured in Router 0. This DNS server has the IP 31.0.0.2.



2.7 Static Routing Configuration

After configuring all router interfaces, static routing was implemented to establish communication between the Routers. Static routes were manually added to each router so that it can correctly forward packets to all remote campus LANs. Each routing entry specifies the destination network, subnet mask and next hop IP address.

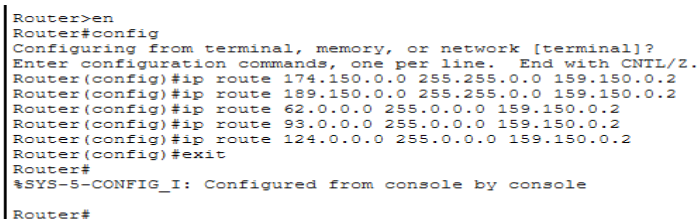
Static routing was chosen for this network design because it offers:

- Complete control over routing decisions
- Predictable and consistent packet forwarding
- Simplicity and transparency, making it suitable for academic analysis and demonstration

This routing approach ensures reliable while keeping the network configuration easy to understand, manage and troubleshoot.

➤ Router0 (Static Configuration)

```
Router>en
Router#config
Router(config)#ip route 174.150.0.0 255.255.0.0 159.150.0.2
Router(config)#ip route 189.150.0.0 255.255.0.0 159.150.0.2
Router(config)#ip route 62.0.0.0 255.0.0.0 159.150.0.2
Router(config)#ip route 93.0.0.0 255.0.0.0 159.150.0.2
Router(config)#ip route 124.0.0.0 255.0.0.0 159.150.0.2
Router(config)#exit
Router#
```



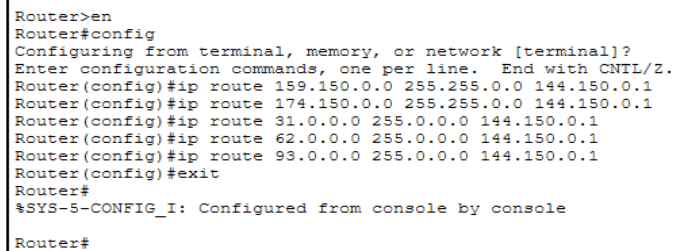
```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 174.150.0.0 255.255.0.0 159.150.0.2
Router(config)#ip route 189.150.0.0 255.255.0.0 159.150.0.2
Router(config)#ip route 62.0.0.0 255.0.0.0 159.150.0.2
Router(config)#ip route 93.0.0.0 255.0.0.0 159.150.0.2
Router(config)#ip route 124.0.0.0 255.0.0.0 159.150.0.2
Router(config)#exit
Router#
SYS-5-CONFIG_I: Configured from console by console
Router#
```

Figure 19: Static Router Configuration for Router 0

➤ Router1 (Static Configuration)

```
Router>en
```

```
Router#config
Router(config)#ip route 159.150.0.0 255.255.0.0 144.150.0.1
Router(config)#ip route 174.150.0.0 255.255.0.0 144.150.0.1
Router(config)#ip route 31.0.0.0 255.0.0.0 144.150.0.1
Router(config)#ip route 62.0.0.0 255.0.0.0 144.150.0.1
Router(config)#ip route 93.0.0.0 255.0.0.0 144.150.0.1
Router(config)#exit
```



```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 159.150.0.0 255.255.0.0 144.150.0.1
Router(config)#ip route 174.150.0.0 255.255.0.0 144.150.0.1
Router(config)#ip route 31.0.0.0 255.0.0.0 144.150.0.1
Router(config)#ip route 62.0.0.0 255.0.0.0 144.150.0.1
Router(config)#ip route 93.0.0.0 255.0.0.0 144.150.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

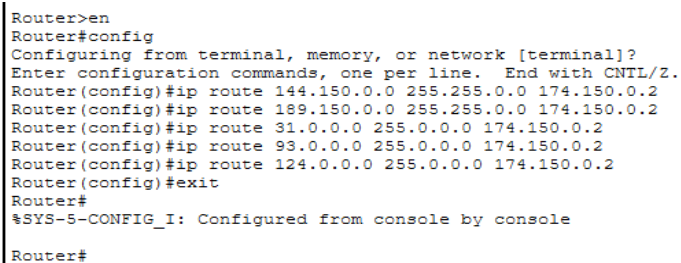
Copy

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Figure 20: Static Router Configuration for Router 1

➤ Router2 (Static Configuration)

```
Router>en
Router#config
Router(config)#ip route 144.150.0.0 255.255.0.0 174.150.0.2
Router(config)#ip route 189.150.0.0 255.255.0.0 174.150.0.2
Router(config)#ip route 31.0.0.0 255.0.0.0 174.150.0.2
Router(config)#ip route 93.0.0.0 255.0.0.0 174.150.0.2
Router(config)#ip route 124.0.0.0 255.0.0.0 174.150.0.2
Router(config)#exit
```



```
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 144.150.0.0 255.255.0.0 174.150.0.2
Router(config)#ip route 189.150.0.0 255.255.0.0 174.150.0.2
Router(config)#ip route 31.0.0.0 255.0.0.0 174.150.0.2
Router(config)#ip route 93.0.0.0 255.0.0.0 174.150.0.2
Router(config)#ip route 124.0.0.0 255.0.0.0 174.150.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

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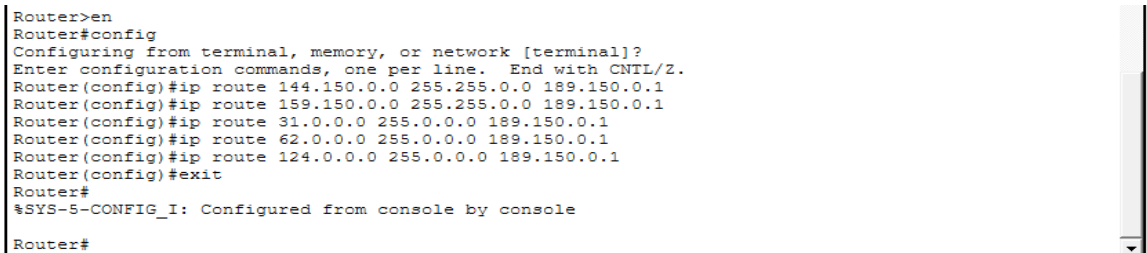
Figure 21: Static Router Configuration for Router 2

➤ Router3(Static Configuration)

```

Router>en
Router#config
Router(config)#ip route 144.150.0.0 255.255.0.0 189.150.0.1
Router(config)#ip route 159.150.0.0 255.255.0.0 189.150.0.1
Router(config)#ip route 31.0.0.0 255.0.0.0 189.150.0.1
Router(config)#ip route 62.0.0.0 255.0.0.0 189.150.0.1
Router(config)#ip route 124.0.0.0 255.0.0.0 189.150.0.1
Router(config)#exit

```



```

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 144.150.0.0 255.255.0.0 189.150.0.1
Router(config)#ip route 159.150.0.0 255.255.0.0 189.150.0.1
Router(config)#ip route 31.0.0.0 255.0.0.0 189.150.0.1
Router(config)#ip route 62.0.0.0 255.0.0.0 189.150.0.1
Router(config)#ip route 124.0.0.0 255.0.0.0 189.150.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#

```

Figure 22: Static Router Configuration for Router 3

3. Experimental Results

3.1 Ping Command

Ping commands were executed between hosts of different LANS (PC11 to PC7 and PC1 to PC4.). Successful ICMP replies confirmed:

- Correct IP addressing
- Proper static routing
- End to end connectivity

➤ Ping Command

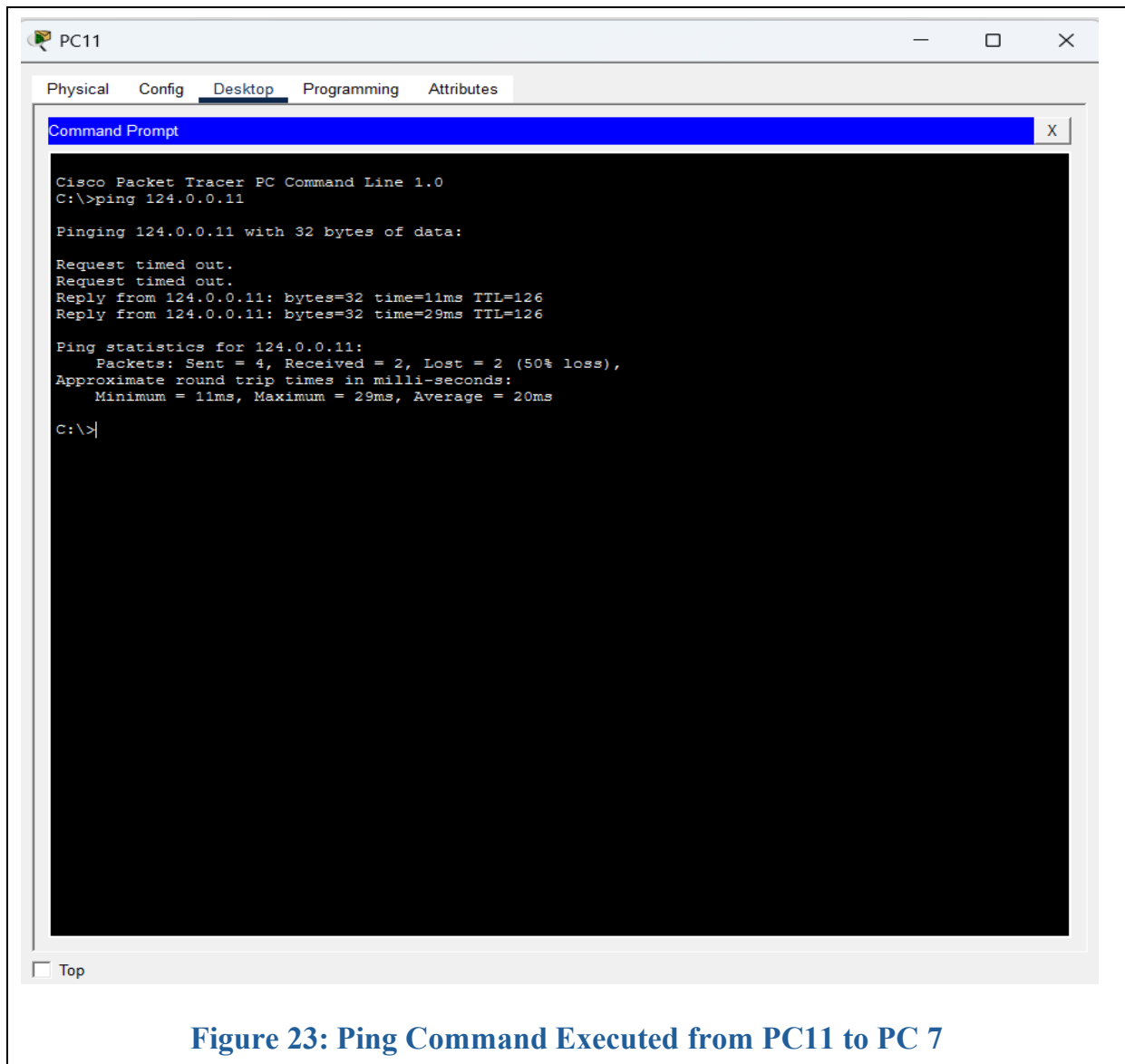


Figure 23: Ping Command Executed from PC11 to PC 7

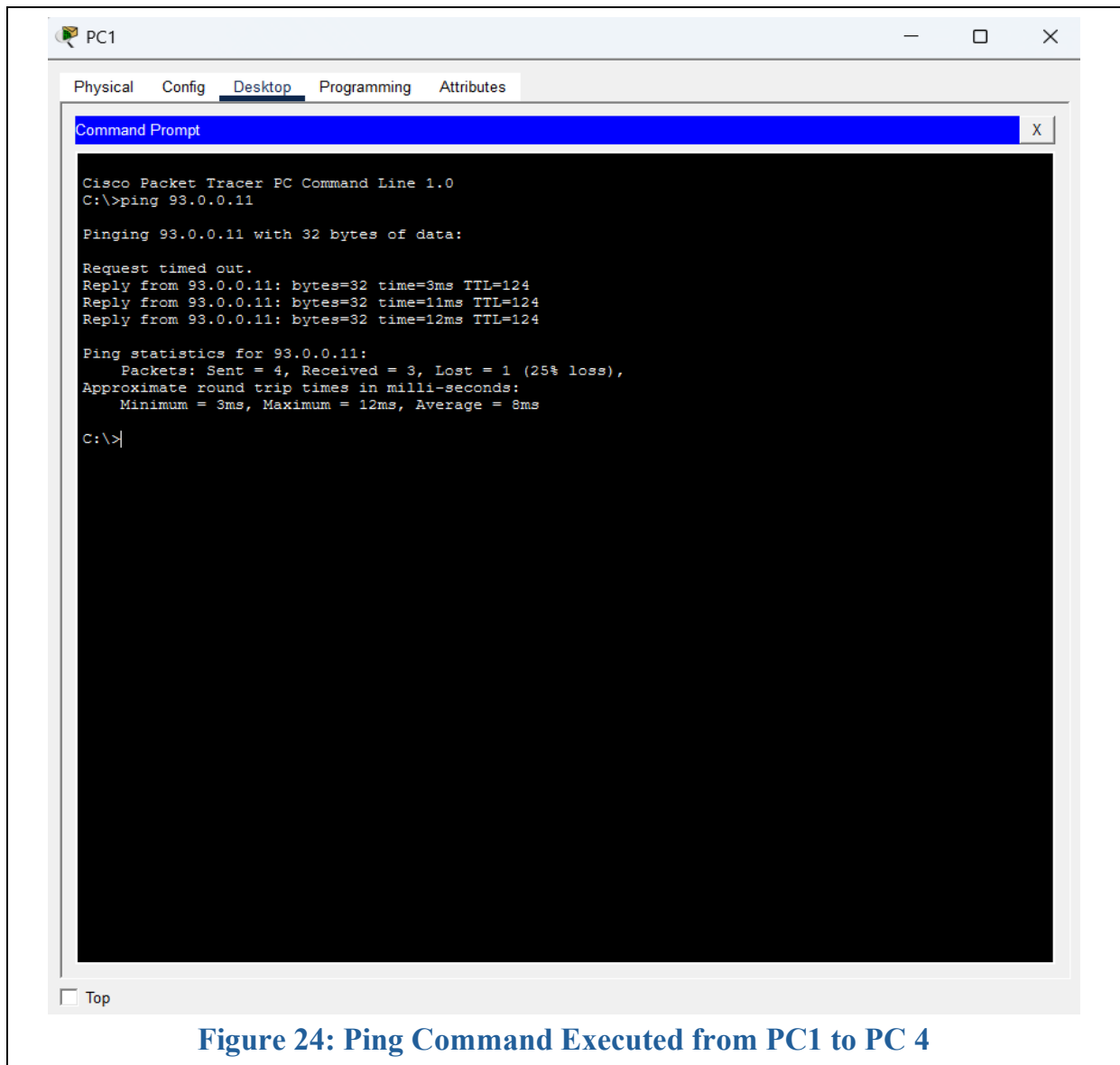
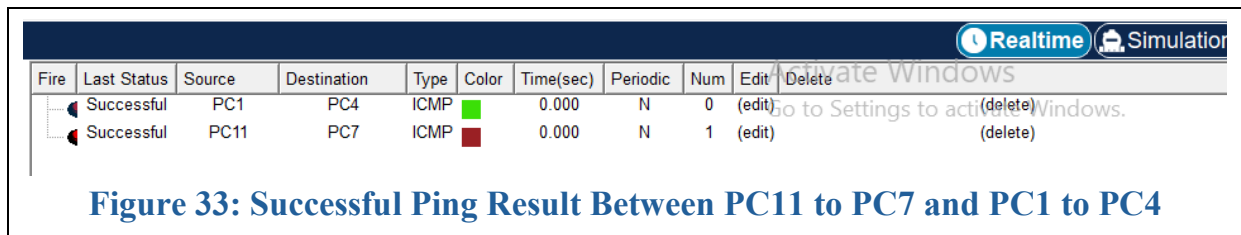


Figure 24: Ping Command Executed from PC1 to PC 4

This figure shows the successful ping results between PC11 and PC7, as well as between PC1 and PC4. The successful replies confirm that IP addressing and subnetting have been configured correctly and that DHCP services are working as expected in each LAN. It also verifies that static routing on all four routers has been properly configured, allowing packets to travel across different networks without any loss. Overall, the ping results demonstrate full end-to-end connectivity and the correct operation of the designed network

3.2 Ping Test Results



4. Conclusion

Here, a complete network was successfully designed and implemented using four routers based on the given topology. Proper network planning was followed to ensure unique addressing and smooth communication between different LANs and WAN links. The use of Class A and Class B IP addresses helped maintain clarity and simplicity in the addressing scheme.

The network was configured in a structured manner so that hosts could communicate across multiple networks without interruption. Router configurations and static routing were carefully applied to ensure reliable data forwarding between all connected networks. The successful communication observed during testing shows that the overall network design and configuration are stable and well organized.

This implementation demonstrates a clear understanding of network design concepts and provides a practical example of how routing and addressing work together in a multi-router environment.